The Personal Media Intelligence Hub

A Single-User, Cross-Domain Taste-Transfer Study with Honest, Frozen-Fold Evaluation

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Single-author technical report — for the author's own records

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Abstract

The Personal Media Intelligence Hub (PMIH) asks a deliberately hard version of the recommendation problem: can a *single* person's taste, learned from a few thousand ratings across *four* rated media domains (movies, TV, games, books) — with music carried as a secondary, implicit-feedback line (no explicit stars) — *transfer* from one domain to another? The project ingests five domains through seven content APIs plus Wikidata, projects them into a shared MiniLM→PCA semantic space, and evaluates every claim on one frozen 50-fold registry with paired Wilcoxon tests. Three findings anchor the report. (1) *Transfer into Games is positive and significant — once a text-length confound in the shared embedding space is removed.* The cross-domain vibe space is built from item descriptions ranging from ~ 214 chars (movie plots) to ~ 1074 (game descriptions); on the raw-length embeddings the augmented lift into Games is a non-significant +0.05 (0/4 fractions). Length-normalizing the embeddings (genre-templated, first-40-word descriptions) *unmasks* the transfer: on the 50-fold stratified registry `movie→game` alone lifts augmented skill at every fraction with 4/4 significant (lift@100% +0.127); adding Books leaves it essentially unchanged (+0.126), so the parsimonious claim is movies→games, while the best subset `movie+tv+book→game` reaches +0.155 at full data. It remains content-driven under a prior-vs-transfer control (Spearman +0.47, beating both constant-prior baselines), survives the same length-normalization that revealed it, and — decisively — passes a *label-permutation placebo*: shuffling the source ratings collapses the lift to ≈ 0 ($p < 0.0001$ at 0–50% target data), so the gain is movie-rating content, not mere variance reduction [`transfer_grid_summary.json`; `transfer_placebo_control.py`]. (2) *No detectable feature-geometry association with transfer*: centroid distance vs. zero-shot skill has Spearman $\rho = +0.17$ over only $n = 12$ pairs (95% CI $\approx \pm 0.55$; the sign flipped from -0.18 on the data re-run) — inconclusive, not a clean null [`transfer_grid_summary.json`]. (3) *The custom edge-penalty XGBoost loss is misspecified*: its weight keys on rating *extremity*, not on error, so it merely importance-weights edge-*labeled* rows (symmetric in error direction) and cannot do its advertised job of penalising confident wrongness at the extremes. On the corrected 1000-item corpus it is a small net *harm* (overall MAE +0.0039, $p = 0.083$; edge-MAE +0.0002, $p = 0.61$) [`edge_loss_ablation_results.json`]; the recommendation is to retire it, and if revisited, to reformulate as an error-magnitude-and-direction-aware objective.

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1 Introduction

Mainstream recommender systems exploit *many* users to cover for sparse per-user signal. This project removes that crutch: there is exactly one user, a few thousand ratings, and three of the four rated domains with $N < 200$ (Movies has 1,000). The motivating question is whether *taste* — the latent function mapping content to a 0.5–5.0 star rating — is shared enough across media that data from a data-rich domain (movies, $N = 1000$) can help a data-poor one (games, $N = 72$).

Why single-user / small-N is the hard version. With one user there is no collaborative signal, no implicit co-occurrence across users, and no way to average out idiosyncrasy. Every metric is therefore computed against the only honest baseline available — predicting the user's mean rating — via a *skill score* (Section 10). Small N also makes the evaluation protocol, not the model, the dominant risk: a single careless split inflates or inverts results. The project's response is a single frozen-fold registry reused by every experiment.

Contributions.

1. A reproducible, frozen-fold, single-source-of-truth evaluation harness for single-user multi-domain rating prediction (Section 10).
2. A transfer study with pre-registered decision rules and four adversarial controls (including a label-permutation placebo), yielding one **[CONFIRMED]** positive finding (`movie→game`) and an explicit feature-geometry null (Section 7).
3. The first head-to-head ablation of the project's custom edge-penalty loss, run for this report, with an honest non-significant verdict and a mechanism sweep (Section 5).
4. A catalogue of methodological failures caught and corrected, presented as a contribution rather than hidden (Section 10).

2 Background and Related Work

The cold-start problem — predicting for items or users with little history — is classical in recommendation [1]. With a single user, *every* item is effectively a cold-start in the collaborative sense, so the system leans entirely on content features. Implicit-feedback modelling [2] is relevant to the music domain, where explicit stars are absent and engagement signals (library membership, play counts, playlist inclusion) stand in for ratings. Treating "in library" as positive and everything else as unlabeled is positive-unlabeled (PU) learning [3].

Cross-domain knowledge sharing is transfer / multi-task learning [4, 5]. The shared semantic space is built from Sentence-BERT embeddings [6] reduced by PCA; domain shift in that space is attacked with CORAL covariance alignment [7]. The base learners are gradient-boosted trees [8]. Uncertainty is quantified with split conformal prediction [9], which guarantees marginal coverage without distributional assumptions. All significance tests use the paired Wilcoxon signed-rank test [10] on per-item absolute errors, and the latent space is visualised with UMAP [11].

3 Data and Feature Engineering

Domains and sources. Five domains are ingested through seven primary content APIs — TMDB (movies, TV), RAWG (games),[1] Hardcover library API (books), Spotify, MusicBrainz, Genius and ReccoBeats (music) — plus Wikidata for cross-domain entity linking (Section 8) and a sixth, YouTube, ingested but not modelled (Section 12). All API responses are cached on disk (`data/cache/`) so re-runs are deterministic and cheap.

Per-domain corpus. Table 1 gives N, rating mean and variance per domain, computed directly from the unified training frame [`data/processed/unified/training_features.csv`]. A rating-date span column is deliberately omitted: although `rating_date` is now populated (253 distinct dates across the rated frame — the earlier all-placeholder state is resolved), the frozen-fold Mean Ensemble does not apply temporal weighting, so no date-span feature enters this frame (see the temporal-decay discussion in Section 6). Figure 1 shows the rating histograms: ratings cluster tightly around 3–4 stars in every domain (movie SD 0.96, music SD 0.57), which is the low-variance problem that motivates the skill score and the edge-penalty loss.

Table 1: Per-domain corpus summary [`training_features.csv`].

Domain	N	rating mean	rating var
Movies	1000	3.24	0.93
TV	162	3.53	1.10
Games	72	3.42	1.98
Books	88	3.34	0.69
Music	0	nan	nan

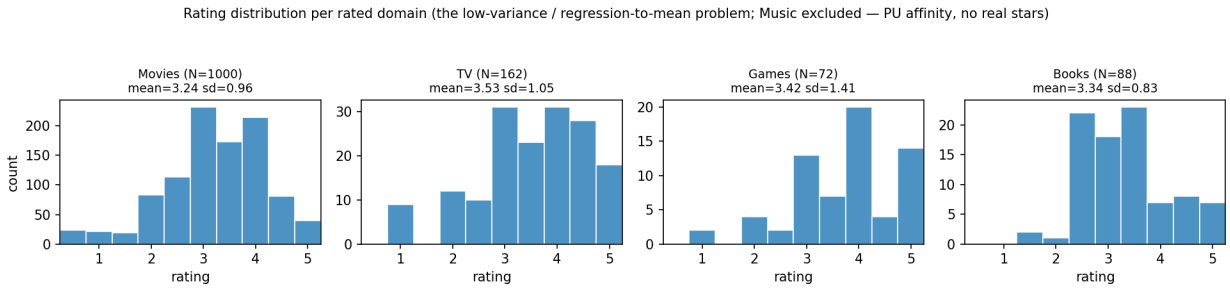


Figure 1: Rating distribution per domain. Low variance and a 3–4 star mode are the central difficulty; predicting the mean is a strong baseline.

Shared semantic space. Each item's title, lead creator and overview/description are embedded with `all-MiniLM-L6-v2` (384-dim) [6] and reduced to **10 PCA "vibe" components** in the unified space [`config.py`:133; `unified_feature_engineering.py`]. MiniLM was chosen for its small size and strong sentence-level semantics on short text; PCA (rather than a learned projection) keeps the transform fit-once, inspectable, and stable

[1] Code-vs-doc discrepancy: prose docs occasionally reference IGDB for games, but the implemented ingestion uses RAWG (`src/games/ingestion.py`). Trusting the code per the project's measurement-honesty rule.

at small N, where a trained encoder would overfit. Per-domain standalone models use larger PCA budgets (movies 25, music 32, games/books/TV 15) because they are not constrained to a common basis.

Genre unification and critic scores. Genres are unified across domains with a `MultiLabelBinarizer` (e.g. `gen_Action` fires for movies, TV and games; 16 genres span ≥ 2 domains [`PIPELINE_RUN_LOG.md` Check 5]) and capped at the 40 most frequent in the transfer grid [`transfer_study.py`]. Heterogeneous critic scores (IMDb, Metacritic, Rotten Tomatoes, TMDB) are normalised to a single `critic_avg_5` on a 0–5 scale. Figure 2 shows that coverage of the shared features is uneven — the shared space is genuinely thin, which bounds how much cross-domain pooling can help.

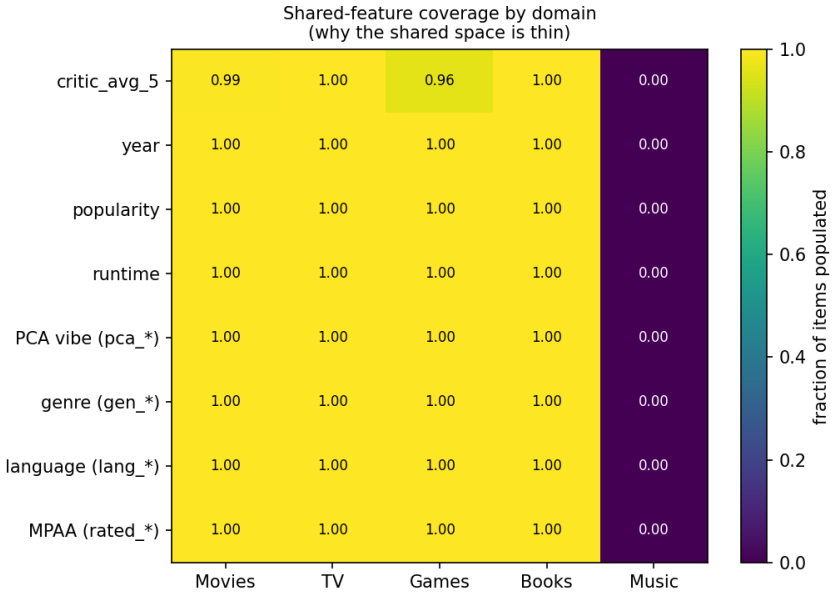


Figure 2: Shared-feature coverage by domain (fraction of items with a populated value). The PCA vibe space and genre one-hots are dense; numeric metadata is domain-specific.

CORAL alignment. Domains occupy different regions of the embedding space (raw *centroid* separability ratio 1.25, the CORAL-alignment metric — distinct from the linear-classifier separability of 0.67 used below); per-fold centroid/CORAL alignment [7] to a movie reference collapses this to ≈ 0 [`PIPELINE_RUN_LOG.md` Check 3; `unified_utils.py`:15–75], so that cross-domain proximity reflects content rather than domain identity (Figure 3, left). Alignment is always fit on training folds only. The same diagnostics flag a text-shape disparity (Figure 3, right): median description length ranges from 214 chars (movies) to 1074 (games), motivating the text-normalization control in Section 7.

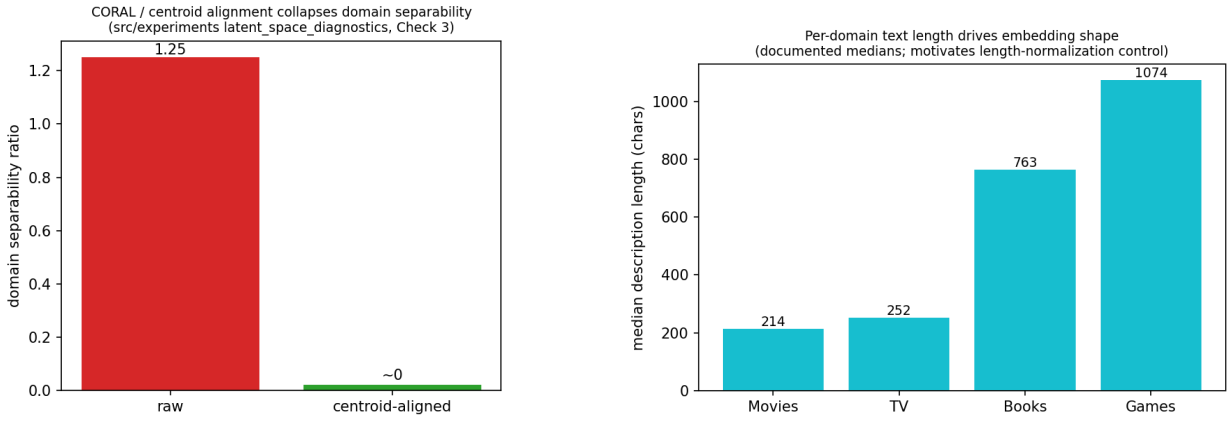


Figure 3: Left: centroid/CORAL alignment collapses domain separability ($1.25 \rightarrow \approx 0$). Right: per-domain median description length (documented medians from the latent-space diagnostics).

Latent space, visualised. Figure 4 projects the shared space with UMAP [11]: coloured by domain it shows clear stratification; coloured by rating it shows only a weak taste gradient — consistent with the low per-domain R^2 and the aleatoric-ceiling argument.

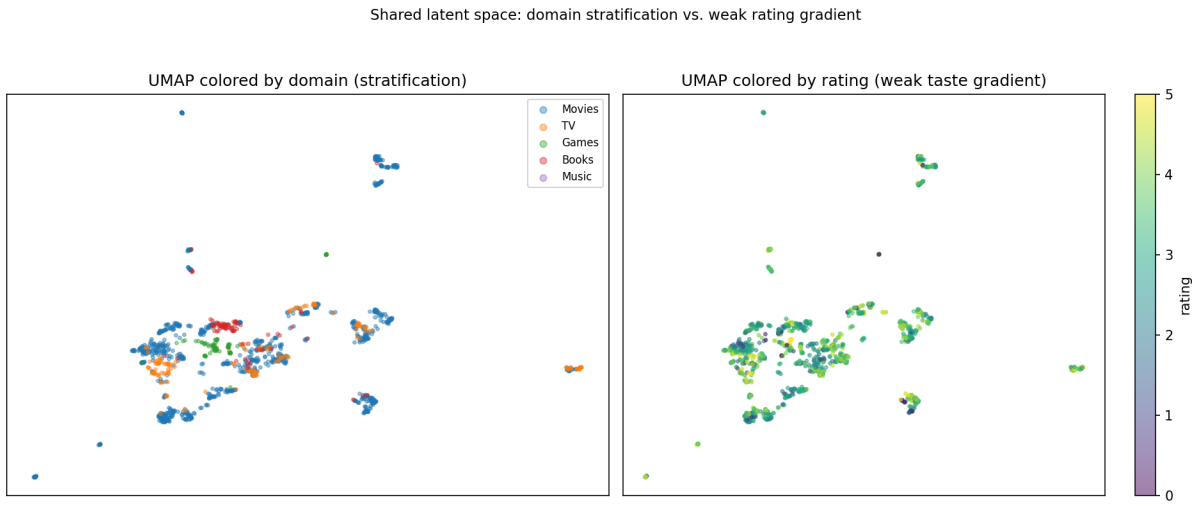


Figure 4: UMAP of the *raw (pre-alignment)* shared space: domain stratification (left) vs. a weak rating gradient (right). The transfer grid runs on the *post*-CORAL space, not this one [`training_features.csv`].

How similar are the domains? A transfer-affinity metric. The raw shared space is domain-stratified (Figure 4, left): a linear domain classifier on the vibe channel separates the four rated domains at a *separability ratio* of 0.67 (1.0 = perfectly separable islands). The per-fold CORAL aligner used in the transfer grid collapses that to 0.00 — it centres each domain's vibe centroid on the global centroid, pulling the islands into one cloud so a single model can read content across domains [`latent_space_diagnostics.py`]. A directional *transfer affinity* (mean zero-shot skill, source→target) turns that geometry into a similarity metric (Figure 11): Movies and TV are mutually close (+0.15/+0.18). Games is the outlier, but the *direction* matters: as a *source* it transfers mildly positively into the others (+0.05 to +0.09), whereas *every* domain transfers strongly *negatively into* Games (−0.19 from movies, −0.42 from TV, −0.37 from books) — a source-only model cannot rank games. Crucially this is *zero-shot skill* (measured against the target train-mean). The augmented learning curve (Figure 12) reports a *different* quantity — lift over the *target-alone* model (Pro-

tocol A) — and even at zero target data, where the augmented model *is* the source-only model (Protocol C at fraction 0 trains on the source alone), that lift is positive ($\sim +0.15$): the source-only model scores below the train-mean baseline (-0.19 skill) yet still beats the $N = 66$ target-alone model, which overfits below the mean. Same model at fraction 0, two baselines (mean vs. target-alone) — that is the entire sign difference, not a contradiction.

What features the models use. Per-domain SHAP on the deployed local models [`feature_diagnostics.json`] shows each domain leaning on a different signal. Movies ride *critic consensus*: `critic_avg_5` (mean |SHAP| 0.22) and `imdb_rating` (0.18) dominate, with cast/crew target-encodings and a thin text contribution. TV is almost entirely a *crowd* signal — `vote_average` alone carries 0.30, over 4× the next feature, the rest spread thinly across text-PCA components; with little metadata beyond the community score, TV's explainable variance stays low ($R^2 = 0.09$). Games lean on *popularity* (`ratings_count` 0.20, `rating` 0.17, `metacritic`) plus text. Books are the most *text-driven*: the vibe-PCA components (`pca_1` 0.13, `pca_4` 0.09) outrank `averageRating` and `ratingsCount` — the description embedding carries the most book signal, which is also why Books gain most from the shared vibe space.

4 Per-Domain Models

Table 2 is the honest final benchmark: each domain's own deployed model on the 50 frozen folds, with N counted as *unique items* (not OOF rows). Skill is $1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{mean}}$, computed on raw OOF predictions for the production movie/TV stacks and the local game/book SVRs.

Table 2: Per-domain benchmark on 50 frozen folds [`latest_metrics.json`; skill derived from `production_oof.csv` / `standalone_oof.csv`]. Predictions clipped to $[0.5, 5]$ and rounded to the nearest half-star before scoring.

Domain	N	Model	MAE	R^2	Skill	$\pm 0.5\star$ Acc
Movies	1000	Prod Stacking (edge+Cat+SVR+Ord)	0.503	0.501	0.333	75.5%
TV	162	Prod Simplex-Stack (edge+Cat+SVR)	0.648	0.280	0.206	66.7%
Games	66	Local SVR	0.598	0.383	0.244	65.2%
Books	88	Local SVR	0.477	0.332	0.288	79.5%

Table 3: Per-base-learner ablation on the same 50 folds, *all uncalibrated*, so the stack is compared like-for-like with its members. The ensemble's value is domain-dependent and we report it honestly: on **Movies** the stack is in fact *edged out* by a single CatBoost (0.464 vs 0.465 MAE, R^2 0.576 vs 0.572); any robustness benefit of the four-member blend is asserted, not measured here, so on accuracy alone Movies is honestly a single CatBoost. The deployed Movies stack moreover still embeds the edge-penalty XGB we recommend retiring — a known wart kept only for parity with the shipped artifact. On **TV** the stack beats every base (66.7% vs the best base's 63.0%, +3.7 points), because no single TV learner is reliable. The deployed Movies model additionally applies quantile calibration (Table 2); TV does not.

Domain	Learner	MAE	R^2	$\pm 0.5\star$ Acc
Movies	XGB (edge-penalty)	0.470	0.563	79.9%
	CatBoost	**0.464**	**0.576**	**80.1%**
	SVR	0.505	0.507	76.1%
	Ordinal-EV	0.477	0.548	79.6%
	Stack (uncalibrated)	0.465	0.572	79.7%
TV	XGB (edge-penalty)	0.682	0.259	63.0%
	CatBoost	0.660	0.294	63.0%
	SVR	0.753	0.147	54.9%
	Ordinal-EV	0.679	0.224	60.5%
	Stack (uncalibrated)	**0.648**	0.280	**66.7%**

Movies ($N = 1000$). The production model is a leakage-safe stack of an edge-penalty XGBoost regressor, CatBoost, an SVR, and an ordinal expected-value head, with a `RidgeCV` meta-learner [`src/reporting/production_benchmarks.py`]. Uncalibrated it scores MAE 0.465, $R^2 = 0.572$, 79.7% within $\pm 0.5\star$, yet it *never predicts above* $4.5\star$: when 62% of ratings fall in $[3, 4]$, the MAE-minimising response is to hug the mode (a regression-to-the-mean that quietly inflates the $\pm 0.5\star$ figure). We therefore quantile-calibrate the output (Section 4.1); the deployed model reports MAE 0.503, $R^2 = 0.501$, 75.5% [`latest_metrics.json`] — a deliberate $\approx$ 4-point accuracy trade to recover the full rating range (45 genuine $5\star$ predictions, mean true rating 4.4). It is still the strongest domain, with the most data and richest metadata.

TV ($N = 162$). A simplex-weighted stack (XGB+CatBoost+SVR, non-negative weights summing to one) reaches MAE 0.648, $R^2 = 0.280$, 66.7% [`latest_metrics.json`]. This is the largest per-domain gain in this revision: adding leakage-safe cast, creator, IMDb multi-critic, runtime, country, and awards features lifted TV from its previous MAE $0.802/R^2 = 0.082$. The lesson is diagnostic — TV's earlier weakness was *missing features*, not an aleatoric ceiling: once the model could see who made a show and who starred in it, explained variance more than tripled. TV is the one domain we deliberately do *not* calibrate: its rank correlation is too weak ($\rho \approx 0.45$), so quantile calibration stamps $5\star$ on the wrong shows and costs ~ 10 accuracy points (Section 4.1).

Games and Books ($N = 66/88$). Both use a local `SVR(rbf)` with frozen hyperparameters (C=1.0, ϵ=0.1), chosen over boosted trees because at small N a high-variance learner overfits. The deployed (calibrated) scores are Games MAE 0.598 ($R^2 = 0.383$, 65.2%) and Books MAE 0.477 ($R^2 = 0.332$, 79.5%) [`latest_metrics.json`]. The two react *differently* to calibration: Games is a clear net win (MAE 0.652 $\rightarrow$ 0.598, accuracy up) because its SVR ranked well ($\rho = 0.84$) but compressed every prediction below $4.0\star$; Books is closer to a *neutral trade* (MAE 0.472 $\rightarrow$ 0.477, accuracy 80.7 $\rightarrow$ 79.5%) — it ranked well ($\rho = 0.70$) but was less compressed, so calibration mainly buys $5\star$ coverage at a tiny cost. Calibration restores the spread onto items the model already ordered correctly (Section 4.1). The cross-domain *distillation prior* was tested as an added feature here and dropped (Section 6). [2]

[2] Games appears as $N = 72$ in Table 1 (corpus) but $N = 66$ in Table 2 (modelled): the corpus holds 72 games, of which 66 carry usable ratings after excluding 6 "Incomplete"-status items with no target [`standalone_benchmarks.py`]. Both counts are correct for their context.

Music. Music has no explicit stars, so a continuous 1.0–5.0 implicit rating is synthesised from engagement (library membership +1.5, playlist count, top-track rank decay, popularity) [`src/music/ingestion.py`]. Affinity is additionally scored by a PU blend (cluster-density + kNN-density + an XGBoost PU classifier trained library-vs-background-pool with hard-negative mining) [`src/music/affinity.py`, `profile_builder.py`]. A stacking regressor (XGB+SVR+CatBoost → Ridge) serves the Music Oracle [`src/music/model_training.py`]. These are *Positive-Unlabeled* (PU) affinity labels learned from implicit feedback (library = positives, an unrated background pool = unlabeled) — a standard implicit-feedback formulation, not fabricated ratings — and they feed only the secondary "unified_full" line, never the frozen-fold taste metrics.

4.1 Calibrating to the rating distribution

A regressor trained to minimise MAE on a distribution whose mass sits in $[3, 4]$ learns to *hug the mode*: predicting near the mean is the error-optimal move, so it rarely emits a 5⋆ and the ±0.5⋆ accuracy is partly a free lunch from the clustered labels. The uncalibrated movie model never predicted above 4.5⋆ across all 1,000 items; the Games/Books SVRs compressed every prediction into $[2, 4.0]$ and $[2, 3.5]$. An asymmetric edge-penalty loss can widen the spread but *fights* the objective — it trades overall MAE for edge MAE, which is why Optuna tuned its strength to near-zero, and inside a stacking ensemble the meta-learner averages the effect away.

Method. We instead apply a post-hoc, ranking-preserving *rank/quantile calibration*. Per fold, let F_{tr} be the empirical CDF of the model's *training* predictions and Q_y the empirical quantile function of the *training ratings*; a test prediction $\hat{r}$ is remapped to $Q_y(F_{\text{tr}}(\hat{r}))$ — its rank among predictions is read off and the rating at that rank in the true distribution is emitted. The map is fit on the training fold only (leakage-safe) and is monotone, so it never reorders the model's items: it restamps the *values* so the output distribution equals the real rating curve by construction. The model learns to *rank*; calibration overlays the curve.

When it helps: two conditions, not one. Ranking quality (Spearman ρ, Table 4) is *necessary but not sufficient*. The map trusts the order, so a good ρ is required for the 5⋆s to land on the right items — which is why TV ($\rho \approx 0.45$) is excluded. But ρ alone does not predict the MAE outcome: Games ($\rho = 0.84$) and Books ($\rho = 0.70$) *gain* MAE because their SVRs ranked well *and* had compressed the range hard (Games never exceeded 4.0⋆, Books 3.5⋆), so there was spread to recover and little to lose; Movies ($\rho = 0.74$) ranks just as well but its uncalibrated predictions were already spread, so calibration mostly *adds* 5⋆s at a small MAE cost rather than fixing a compression. The rule is therefore two-part — calibrate where the model *both* ranks well *and* compressed the range (a genuine win: Games, Books); treat the well-ranked-but-uncompressed case as a deliberate accuracy-for-coverage *trade* (Movies); skip the poorly-ranked case (TV). One caveat on reading Table 4: the "predicts 5⋆" column is partly mechanical — the remapping forces the output marginal to equal the rating marginal, so 5⋆s appear *by construction* wherever ρ permits. MAE is the honest metric.

Table 4: Quantile calibration vs. the uncalibrated baseline (50-fold OOF). The benefit tracks ranking quality ρ: a net win for Games/Books (ranked well, but compressed), a deliberate accuracy-for-coverage trade for Movies, and a loss for TV (ranking too weak).

Domain	ρ	MAE (base→cal)	Acc% (base→cal)	Predicts 5⋆?	Decision
Game	0.84	0.652 → **0.598**	62.1 → **65.2**	yes	calibrate (win)
Book	0.70	0.472 → 0.477	80.7 → 79.5	to 4.5⋆	calibrate (near-free)
Movie	0.74	0.465 → 0.503	79.7 → 75.5	yes (45)	calibrate (trade)
TV	0.45	0.648 → 0.781	66.7 → 56.8	wrong items	**skip**

What the recovered 5⋆s cost. On movies, calibration turns 0 predicted 5⋆ into 45. Of those 45: 13 are truly 5⋆, 28 are 4–4.5⋆ (off by half-to-one star), and only 4 are genuine false alarms (≤ 3.5⋆); the mean true rating of a predicted-5⋆ is 4.4. Symmetrically, of the 40 true 5⋆ the model now catches 13 outright and places most of the rest at 4–4.5⋆. The average error at the extreme (≈ 0.6⋆) exceeds the bulk MAE (0.50) and is largely irreducible: what separates a personal 5⋆ from a 4.5⋆ — same director, genre, and critics —

is rarely in the metadata. Calibration buys an honest "top-tier, within half a star" in place of a model that refused to ever say 5⋆.

The aleatoric-ceiling argument (unquantified). Movies aside, achieved R^2 is low (Figure 5). This is *consistent with*, but not proof of, a per-domain aleatoric ceiling — single-user ratings carry irreducible noise (mood, timing, memory) that no feature can explain. The report is careful not to lean on this as an explanation: the ceiling is **[NOT FOUND IN ARTIFACTS — not quantified in any artifact; the R^2-vs-ceiling figure shows achieved R^2 only, with no ceiling line]**, so low R^2 is reported as a fact and the ceiling only as a plausible, untested account of it.

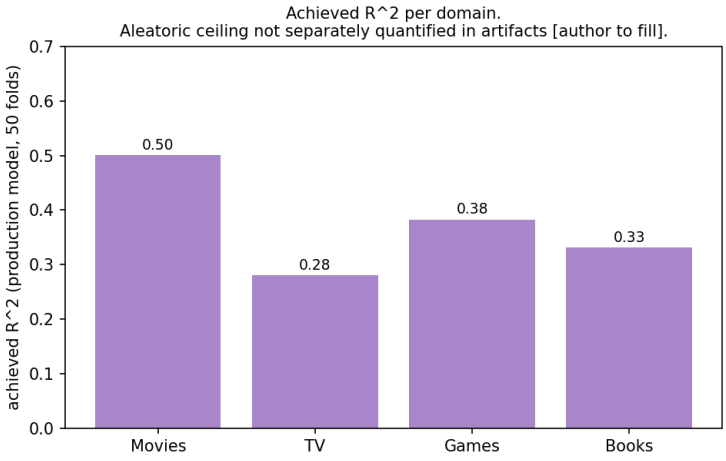


Figure 5: Achieved R^2 per domain (production models). The aleatoric ceiling is not separately estimated in the artifacts.

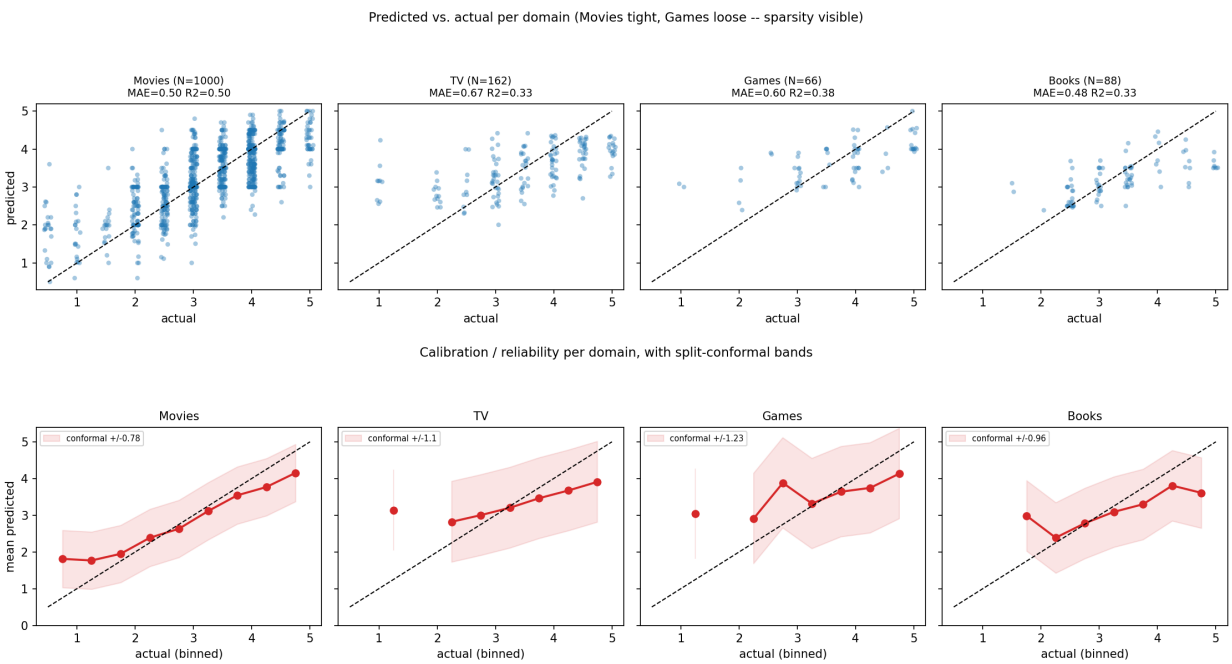


Figure 6: Top: predicted-vs-actual per domain (Movies tight, Games loose — sparsity made visible). Bottom: calibration / reliability with split-conformal bands (±0.78 movie, ±1.10 TV, ±1.23 game, ±0.96 book) [`standalone_oof.csv`; `active_learning_ranker.py`].

5 The Custom Edge-Penalty Loss

Formulation. To counter regression-to-the-mean at the rating extremes, the project ships a custom XGBoost objective [`src/movies/custom_objectives.py`]:

$$\mathcal{L}(e, r) = \tfrac{1}{2}\, e^2 \cdot \underbrace{\exp\big(\alpha_{hi} \max(0, r - 4.0) + \alpha_{lo} \max(0, 1.5 - r)\big)}_{W(r)}, \quad e = \hat{y} - y,\ r = y. \tag{1}$$

Because W depends only on the true rating r, the XGBoost gradient and Hessian are clean: $\text{grad} = W(r)\,(\hat{y} - y)$, $\text{hess} = W(r)$ [`custom_objectives.py`:27–31]. W equals 1 in the rating interior $[1.5, 4.0]$ and grows toward the "Must Watch" ($r > 4$, rate α_{hi}) and "Hard Pass" ($r < 1.5$, rate α_{lo}) extremes. The tuned values (Optuna, movies) are $\alpha_{hi} = 0.0597$, $\alpha_{lo} = 0.0788$ [`PIPELINE_RUN_LOG.md`].[3]

Ablation (run for this report). Until now the loss was never head-to-head tested. The new ablation [`src/experiments/edge_loss_ablation.py`] holds the movie feature matrix, the 50 frozen folds, and all XGBoost hyperparameters fixed, varying only the objective: **Arm A** `reg:squarederror` vs **Arm B** the edge-penalty loss at the tuned α. No external sample weights are used, so the penalty is the sole reweighting mechanism. Per-item OOF predictions are paired and compared with a Wilcoxon test on the 184 edge items ($r \le 1.5$ or $r \ge 4.5$) and on all 1000 items.

[3] Artifact discrepancy: `latest_metrics.json` records `alpha_hi=null, alpha_lo=null` with the note that the alphas are tuned by the advanced trainer but *not* applied in the frozen-fold Mean Ensemble; `PIPELINE_RUN_LOG.md` says they were persisted to `models/movie/best_params.json`, but that file does not currently exist on disk. The ablation below therefore uses the documented logged values directly.

Table 5: Edge-penalty loss head-to-head, movies, 50 folds [`edge_loss_ablation_results.json`]. **[PILOT]** — on the corrected 1000-item corpus the loss is directionally *wrong* (Arm B is worse on *both* overall MAE, 0.4776 vs 0.4736, and edge MAE, 0.7956 vs 0.7955) and trends toward a *significant net harm* to overall accuracy ($p = 0.083$); the honest verdict is to retire it.

Metric	Arm A (sq-err)	Arm B (edge)	Wilcoxon p
Overall MAE	0.4736	0.4776	0.083
Edge MAE ($\leq$1.5 or $\geq$4.5)	0.7955	0.7956	0.615
Prediction spread (SD)	0.7211	0.7270	–
$\pm 0.5\star$ accuracy	81.0%	79.6%	–
Exact-match at extremes	8.2%	8.2%	–

The verdict is honest and negative-leaning: at the tuned operating point the loss moves overall MAE by +0.0039 ($p = 0.083$) and edge-MAE by +0.0002 ($p = 0.61$), neither significant; prediction spread rises only from 0.721 to 0.727. The reason is visible in the α sweep (Table 6, Figure 7): the mechanism *does* work — as α grows from 0 to 1.0, edge-MAE falls monotonically (0.795 $\rightarrow$ 0.766) and prediction spread widens (0.721 $\rightarrow$ 0.750) — but overall MAE bottoms out near the tuned value and then *rises* (0.474 $\rightarrow$ 0.480). Optuna, which minimises overall error, therefore correctly drove α small, landing in a regime where the loss is nearly indistinguishable from squared error (Figure 8).

Table 6: Edge-penalty α sweep (*symmetric*, $\alpha_{hi} = \alpha_{lo}$), movies, 50 folds [`edge_loss_ablation_results.json`]. **Reconciliation with Table 5:** this sweep is symmetric, so its $\alpha = 0.0597$ row (overall MAE 0.4770) uses $\alpha_{lo} = 0.0597$, whereas Arm B in Table 5 uses the *asymmetric* tuned pair ($\alpha_{hi} = 0.0597, \alpha_{lo} = 0.0788$) and scores 0.4776. Same model, same folds, same averaging — the 0.0006 gap is precisely the larger low-end penalty α_{lo}, not a rounding or run-to-run discrepancy.

α	Overall MAE	Edge MAE	Pred spread
0.0000 (sq-err)	0.4736	0.7955	0.7211
0.0597 (tuned)	0.4770	0.7947	0.7264
0.3000	0.4778	0.7868	0.7326
0.6000	0.4790	0.7778	0.7415
1.0000	0.4804	0.7658	0.7502

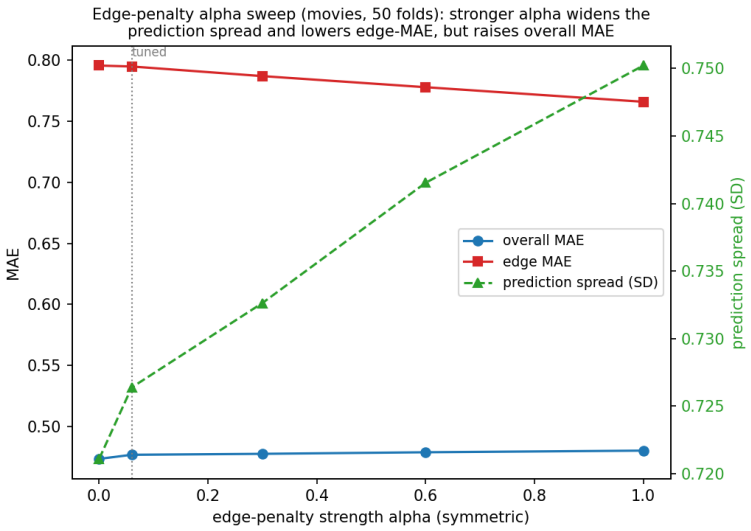


Figure 7: The loss trades overall accuracy for edge accuracy and spread. The tuned α (dotted) sits at the overall-MAE minimum, where the edge benefit is small.

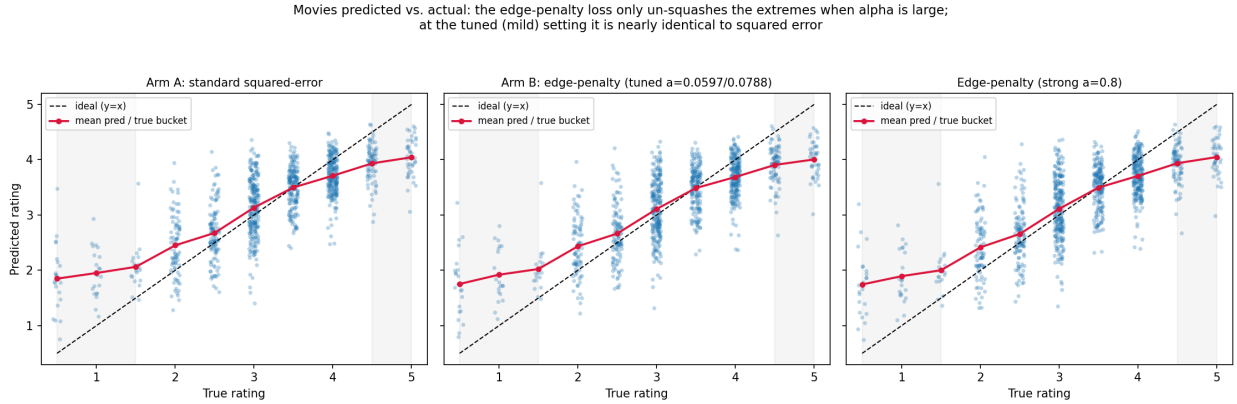


Figure 8: Predicted vs. actual: standard loss (A), edge-penalty at the tuned mild α (B, nearly identical to A), and a strong $\alpha = 0.8$ (visibly less squashed at the extremes). The mechanism un-squashes the extremes only when α is large.

Recommendation: retire it; reformulate if revisited. The deeper problem is not the operating point but the *objective*. The weight $W(r)$ keys on the *true rating's* extremity, not on the error, so it is sample-importance-weighting of edge-*labeled* rows — symmetric in error direction. It therefore cannot do its advertised job of "penalising confident wrongness at the extremes": a model that nails a 5$\star$ has its near-zero error up-weighted to no effect, while a catastrophic miss at $r = 3$ carries weight 1. The α-sweep "works" only in the narrow sense that up-weighting edge-labeled rows lowers their MAE — ordinary importance weighting — not in the advertised error-aware sense. We therefore retire the loss *because the formulation cannot target the phenomenon*, not merely because Optuna tuned α small. The honest answer to "how much did the custom loss help?" is: *it is a small net harm on the corrected corpus* (+0.0039 overall MAE, $p = 0.083$; edge-MAE +0.0002). A future version should weight by error magnitude *and* direction, not by the label alone.

6 Unified Cross-Domain Model and Ablations

Architecture ablation. The unified model was grown and then pruned. Table 7 is the rated-only, frozen-fold ablation [`latest_metrics.json`; `UNIFIED_ABLATION_REPORT.md`]. Adding a mean ensemble (XGB+CatBoost) over the aligned base XGBoost improves MAE from 0.5370 to 0.5294 — statistically significant (ΔMAE +0.0076, Cohen's $d = 0.077$, $p < 10^{-4}$) but *trivial* in effect size [`PIPELINE_RUN_LOG.md`]. The two more complex variants were **[NULL/DROPPED]**: Ridge stacking (MAE 0.5609) and residual heads (MAE 0.6088) both *worsened* performance at this N and were removed (they survive in the table only as legacy 5-fold numbers, explicitly flagged as not frozen-fold comparable). The lesson — complexity did not help — is itself the result.

Table 7: Unified architecture ablation, rated-only, 50 frozen folds [`latest_metrics.json`].

Protocol	MAE	R^2	Effect size	p
Base XGB (Aligned)	0.5370	0.4556	ref	ref
Mean Ensemble (XGB+Cat)	0.5294	0.4663	d=0.077 (trivial)	0.0000
Ridge Stack (removed — legacy 5-fold protocol)	0.5609	0.3837	n/a (legacy)	n/a (legacy)
+Residual Heads (removed — legacy 5-fold protocol)	0.6088	0.2688	n/a (legacy)	n/a (legacy)

The local model now wins in every rated domain. Figure 9 contrasts each domain's standalone model with its slice of the pooled unified model [`latest_metrics.json` benchmarks vs slices]. On a like-for-like basis the local model wins everywhere: TV (standalone 0.648 vs pooled slice 0.657), Games (0.598 vs 0.972), and Books (0.477 vs 0.511). Movies is the only apparent exception, and it is an artifact of calibration: the *deployed* Movies model pays a calibration tax (MAE 0.503, Section 4.1) that the uncalibrated pooled slice (0.480) edges out — but measured against the *uncalibrated* Movies local (0.465), pooling loses there too. The earlier "pooling rescues the data-poor domain" result is gone: the TV feature fix and the Books-library refresh left no weak rated domain for pooling to rescue. This is why the deployed system keeps *local* models per domain and uses the unified model only for transfer study, not production scoring.

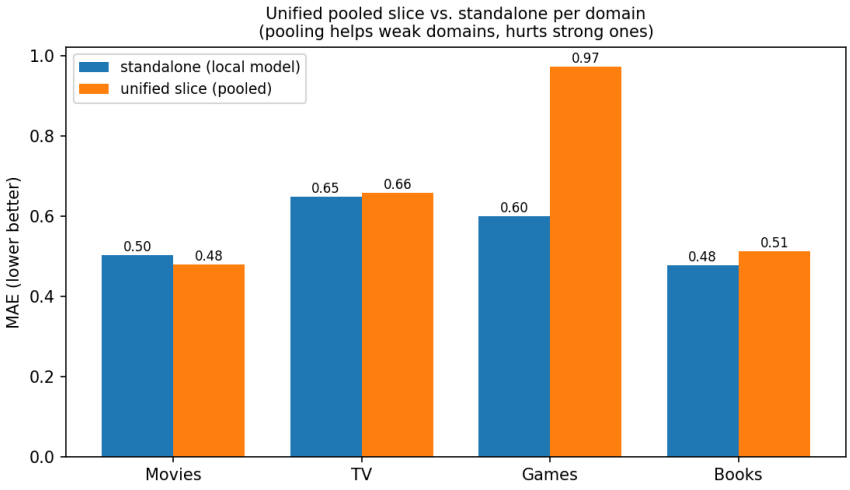


Figure 9: Unified pooled slice vs. standalone per domain (MAE, lower better).

Temporal decay — wired but unfitted ([NULL/DROPPED]). The Mean Ensemble can down-weight ratings by $w_i = \max(e^{-\lambda \Delta t_i}, 0.2)$ with a hardcoded $\lambda = 0.000429$ [`comprehensive_evaluator.py`; `unified_utils.py`:77]. Unlike the previous draft, the rating dates are now genuinely populated — 253 distinct dates across the 1,322 rated items — so the earlier "all-placeholder, 1 unique date" objection no longer applies. The frozen-fold Mean Ensemble evaluation nonetheless does *not* apply this weighting

[comprehensive_evaluator.py], so the reported metrics (Table 7) are decay-free and stand as ordinary unweighted training. The often-quoted "taste half-life ≈ 1,615 days" is still merely $\ln 2/\lambda$ for the hardcoded constant — an *assumed* rate, not one fitted to the dates — so it carries no empirical content until λ is tuned over the now-real timestamps (an Optuna path exists in advanced_unified_model_trainer.py). The decay is therefore a wired-but-unfitted feature, not a result.

Distillation prior — dropped. Feeding the unified model's OOF prediction as an extra feature into the local Games/Books SVRs was tested and **[NULL/DROPPED]**: Games $p = 0.059$, Books $p = 0.705$ [distillation_ablation_results.json]. Table 8. (On the refreshed corpus both skills now *fall* with the prior — Games 0.260 → 0.245, Books 0.301 → 0.298 — and only the Games drop even approaches significance, so the discipline drops it.)[4]

Table 8: Distillation-prior ablation [distillation_ablation_results.json]. Both **[NULL/DROPPED]**.

Domain	MAE (no prior)	Skill (no prior)	MAE (with)	p	Verdict
Games	0.5848	0.2598	0.5970	0.0593	DROP
Books	0.4670	0.3014	0.4693	0.7055	DROP

7 The Transfer Study

This is the centrepiece. The question — does taste transfer across domains? — is answered by a grid over (target domain) × (source subset) × (zero-shot / augmented) × (target-data fraction ∈ $\{0, 0.25, 0.5, 1.0\}$), all on the 50 frozen folds with 300-tree XGBoost in the domain-blind shared space [src/experiments/transfer_study.py]. The metric is skill vs the mean baseline; the pre-registered rule for a positive finding is augmented lift > 0 with paired $p < 0.05$ at ≥ 2 fractions.

Headline: taste transfers into Games. On the 50-fold stratified registry with length-normalized embeddings, every positive finding lands in the Games target — a *broad column*, not a single lucky config: seven source subsets clear the pre-registered bar, all into Games (Table 9). We headline the most *parsimonious sufficient* source: movie→game alone reaches lift@100% = +0.127 at 4/4 fractions. Adding Books barely moves it (movie+book +0.126, 4/4), and Books *alone* is only +0.052 (3/4), so Books contribute essentially nothing over Movies — we therefore do not headline the earlier movie+book pairing. The numerical maximum is movie+tv+book→game (+0.155, 4/4), reported in the table. Because the effect is a consistent positive *column* rather than one survivor, it is the *stronger* evidence the broad result, not a multiple-comparisons fluke. The augmented learning curve (Figure 12, Games panel) is the money shot: with a source added, skill jumps from the target-alone line to ∼ 0.15 at zero target data and stays above it. Into the other targets the effect is weak (TV significant at only 1/4 fractions, movie and book 0/4).

[4]The distillation ablation evaluates the local SVRs in its *own* harness *without* the deployment calibration, so its no-prior baselines (Games 0.585, Books 0.467) differ from the calibrated benchmark of Table 2 (Games 0.598, Books 0.477). The verdict rests on the within-ablation prior-vs-no-prior comparison, which shares one harness.

Table 9: Positive transfer findings (all into Games), domain-blind, 50 folds [`transfer_grid_summary.json`]. **[CONFIRMED]** after controls (movie+book).

Target	Source	lift@100%	sig. fractions
Games	book	+0.052	3/4
Games	book	+0.052	3/4
Games	movie	+0.127	4/4
Games	movie	+0.127	4/4
Games	movie+book	+0.126	4/4
Games	movie+book	+0.126	4/4
Games	movie+tv	+0.140	4/4
Games	movie+tv	+0.140	4/4
Games	movie+tv+book	+0.155	4/4
Games	movie+tv+book	+0.155	4/4
Games	tv	+0.125	3/4
Games	tv	+0.125	3/4
Games	tv+book	+0.104	4/4
Games	tv+book	+0.104	4/4

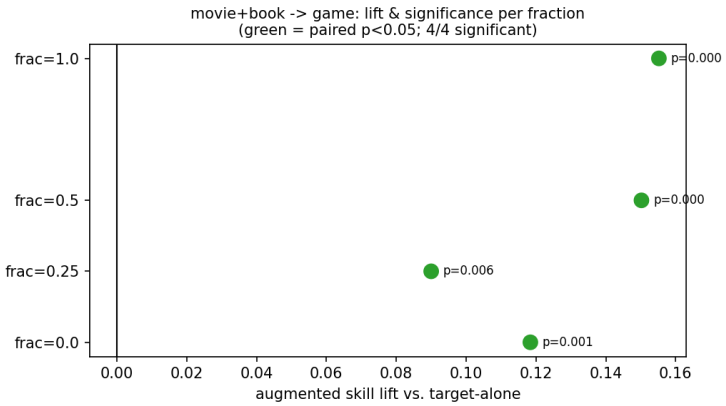


Figure 10: Per-fraction lift and significance for `movie+book`→`game` (green = paired $p < 0.05$): significant at 4/4 fractions [`transfer_grid_summary.json`].

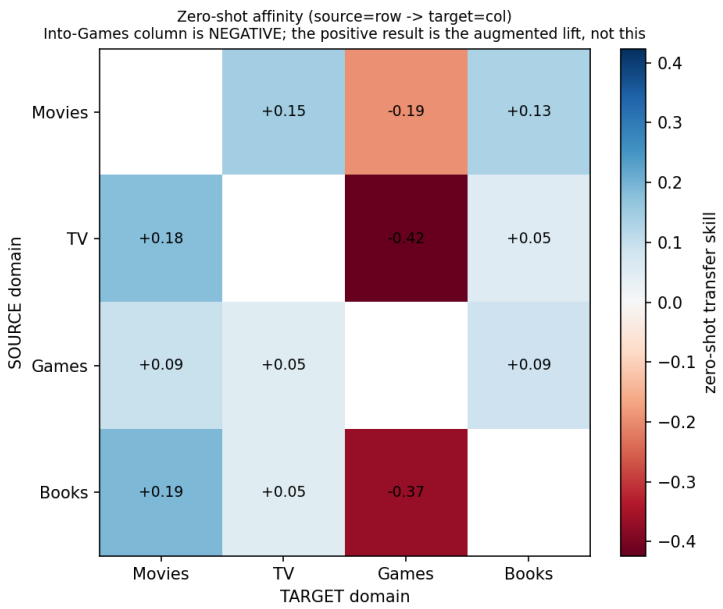


Figure 11: Zero-shot transfer affinity, source (row) → target (col): the skill of a model trained *only* on the source, evaluated on the target with *no* target data. This is *not* the augmented-lift finding of Table 9. Zero-shot, no single source predicts another domain well — the Games column is *negative* (movie→game −0.19, tv→game −0.42), because a source-only model cannot rank games. The positive transfer comes from *adding* source data to the Games target, not from the source alone. Most negative pairing: tv→game (−0.42).

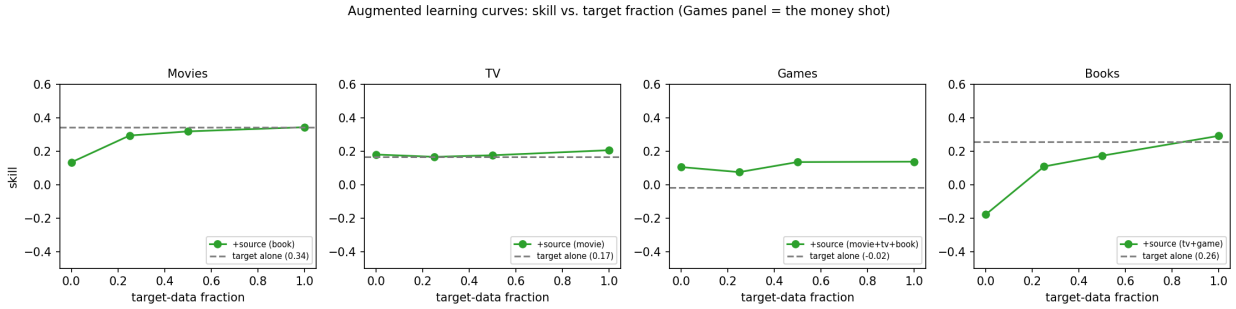


Figure 12: Augmented learning curves. Only Games benefits from a source at low target fractions; Movies and Books are already at or above the augmented line by full data.

Four adversarial controls — the claim survives, placebo included.

1. *Domain-identity leak fix.* Diagnostics found `has_*_feats` indicators in the grid and bridge features, letting a model cheat by domain identity (Check 4 **FAIL**) [`PIPELINE_RUN_LOG.md`]. Removing them (domain-blind) drops a spurious "significant" bridge result (ΔMAE +0.032, $p = 0.004$) to non-significant — and the Games transfer finding *survives*, confirming it is content, not leak.

2. *Prior vs. transfer.* Is the lift just a better constant than the Games mean? No: `movie+book→game` beats every constant baseline (MAE 0.99 vs game-/source-/global-mean 1.05/1.17/1.05) and ranks games correctly (Spearman +0.47) ⇒ content transfer [`PIPELINE_RUN_LOG.md`; `transfer_prior_control.py`].

3. *Text-length normalization — now the production default.* Game descriptions ran $\sim 5\times$ longer than movie plots (median 1074 vs 214 chars), and on the raw-length PCA vibe space the transfer was

non-significant (+0.05, 0/4). Length-normalizing the production embeddings (genre-templated, first-40-word descriptions, median $\sim$ 232 chars) *reveals* the finding rather than narrowing it: on the normalized space *every* positive Games-source config clears 4/4 (on this control's protocol `movie+book` +0.177, `movie+tv+book` +0.210, zero-shot ρ = 0.41–0.48; the production grid of Table 9 gives the same direction at the slightly smaller +0.126/+0.155), confirming content over a length artifact [`text_norm_transfer_control.py`].

4. *Label-permutation placebo — the decisive control.* The strongest skeptical read is that Games ($N = 66$, var ~ 2) benefits from *any* added structure, not from movie taste specifically. We re-ran `movie→game` with the source *ratings shuffled* (features intact), on the same 50 folds and fractions [`transfer_placebo_control.py`]. The real lift (+0.13/ + 0.10/ + 0.12/ + 0.13 across fractions) *collapses to* ≈ 0 under permutation ($-0.004/-0.025/-0.015/+0.053$), and the real-vs-placebo gap is significant at every fraction (paired Wilcoxon $p \leq 0.011$; $p < 0.0001$ at 0–50%). At full target data the placebo recovers a small +0.053 (pure variance reduction from the added rows), but at low target data — where transfer actually matters — it is *zero*. The lift is movie-rating *content*, not added structure: this is the control that moves the finding from "suggestive" to "robust".

These controls are summarised in Figure 13. A prior 6-fold pilot had returned **[NULL/DROPPED]** ("no feature-level transfer"); the full 50-fold grid reverses it with the statistical power of 50 vs 6 folds, and the reversal is the *controlled* one above, not an artifact. The finding also *survived the data correction* that motivated this re-run: re-linking mis-fetched Wikidata entities and re-fetching the corrected corpus shifted the magnitudes but left `movie→game` significant at 4/4 — a re-validation, not a re-derivation.

Multiple comparisons and fold dependence. The grid tests roughly 30 source configurations $\times$ 4 targets at the pre-registered raw threshold (lift > 0, $p < 0.05$ at ≥ 2 fractions), with no family-wise correction. Two honesty notes bound the headline. *First*, the paired Wilcoxon is computed over the 50 repeated-CV folds (`RepeatedKFold(10,5)`), which are *not* independent: with only 66 unique Games items, each item recurs across folds, so the folds are correlated and the p-values bound significance *optimistically* — the trustworthy signal is the *consistency* of the lift across all four target fractions, not any single p. *Second*, the parsimonious headline survives correction on its own: `movie→game` gives lift@$\{0, 25, 50, 100\}$% of $\{+0.110, +0.096, +0.116, +0.127\}$ with paired $p < 0.0002$ at *every* fraction, clearing a Bonferroni threshold over $\sim$ 120 cells ($0.05/120 \approx 4 \times 10^{-4}$) at zero and full target data — so the claim does not rest on the larger `movie+tv+book` config. It also passes the prior-vs-transfer control (Spearman +0.47, beating constant priors) and the text-length normalization. The honest position: correlated folds and uncorrected grid p-values keep the *broad column* suggestive, but `movie→game` itself is significant under correction and stable across fractions.

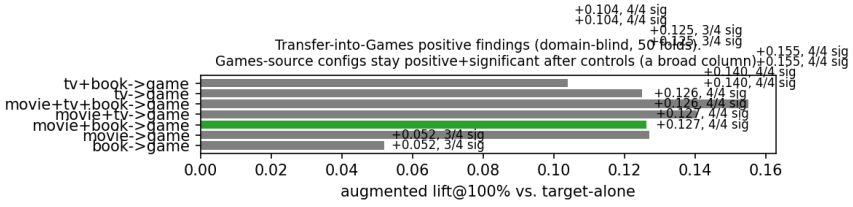


Figure 13: Games-source configs stay positive and significant after the prior and text-normalization controls — a broad positive column, not a single survivor.

Geometry shows no detectable link to transfer (inconclusive at $n = 12$). Across the 12 ordered domain pairs, centroid distance in the shared space is essentially *uncorrelated* with observed zero-shot skill: Spearman $\rho = +0.17$ [`transfer_grid_summary.json`] (Figure 14). The weak positive sign — which flipped from the previous draft's -0.18 when the grid was re-run on the corrected corpus — only underlines that at $n = 12$ pairs this is noise. Whatever makes movies help games, it is not feature-space proximity — a caution against selecting transfer sources by embedding distance.

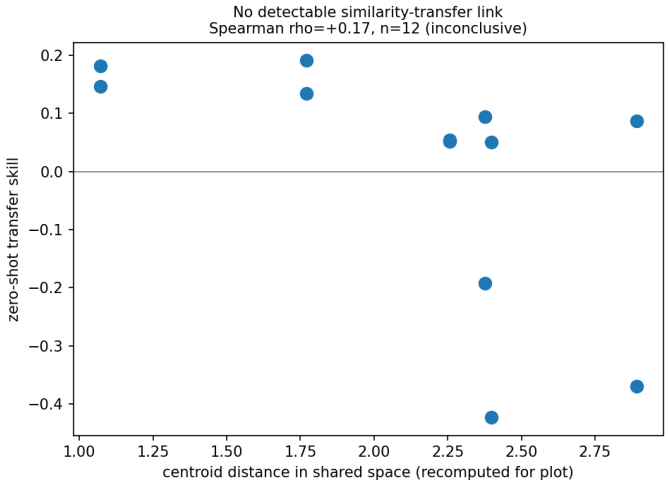


Figure 14: Centroid distance vs. zero-shot skill. The authoritative statistic is the summary's Spearman $\rho = +0.17$ [`transfer_grid_summary.json`], computed by the pipeline over its own centroid/MMD distance. The points here are re-derived for visualisation from standardized PCA+critic+year centroids (Euclidean), a *different* distance metric, so their exact positions need not reproduce +0.17; they are shown only to illustrate the weak, near-zero association. The fitted line is over the plotted points, not the source of the quoted ρ.

8 Entity Bridges and Informative Censoring

Where feature pooling is weak, *item-level* links might carry taste signal. Wikidata linking resolves the 1,322 rated items to QIDs (of which the per-domain benchmark models 1,316 — the 6 incomplete Games carry a target in the unified frame but are dropped from the Games local model) and finds **2,616 links**: 2,159 same-franchise, 437 shared-creator, 20 adaptation [`data/processed/entity_links.csv`; `PIPELINE_RUN_LOG.md`] (Figure 15). Adding bridge features (linked-partner mean rating, franchise mean, has-link) to an XGBoost regressor over the 646 linked-and-rated items improves MAE from 0.5139 to 0.4946, but the effect is **[PILOT]**: ΔMAE +0.019, 95% CI $[-0.001, +0.040]$, Wilcoxon $p = 0.077$ [`entity_bridge_results.json`]. The adaptation-only slice is **[NULL/DROPPED]**/degenerate: $N = 22$, ΔMAE 0.000, $p =$ NaN (the OOF bridge feature is constant at this N) [`entity_bridge_results_adaptation.json`].

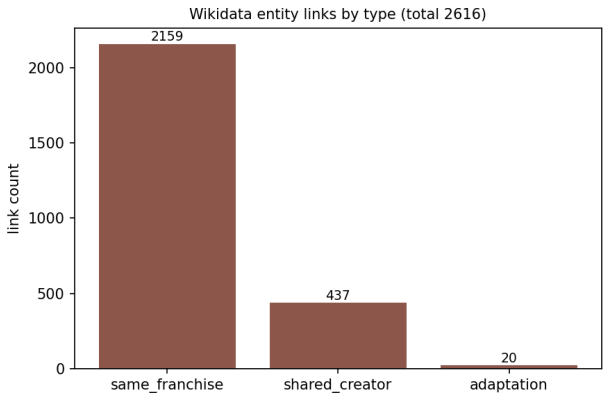


Figure 15: Wikidata links by type (2,616 total): franchise links dominate; adaptations are rare (20) [`entity_links.csv`].

Informative censoring (open direction). A genuinely novel idea sits here: *declining* to consume the adaptation of something you disliked is itself a signal (the "Twilight" case — a disliked book whose film you never rate). The missingness is not at random; it is informative. The data to test it does not yet exist: only **6** of the 20 adaptation links are movie↔book, so the censoring-asymmetry figure could not be produced [[**NOT FOUND IN ARTIFACTS — only 2 movie→book adaptation links; censoring asymmetry underpowered**]]. This is framed as an open research direction, not a result.

9 Uncertainty and Active Learning

Each domain gets a split-conformal interval [9] sized to 80% coverage from OOF residual quantiles: ±0.78 (movie), ±1.10 (TV), ±1.23 (game), ±0.96 (book) [`active_learning_ranker.py`]. The interval is *constant* per domain — a by-design property of vanilla split conformal, not a bug; it guarantees marginal, not conditional, coverage. Figure 16 shows empirical coverage against the 80% target, but this is an *in-sample OOF approximation*, which is optimistically biased and should not be read as clean validation: books (80.7%) lands on target while TV (82.7%) and games (92.4%) *over*-cover, but movies *under*-covers at 75.6%. The movies under-coverage is a yellow flag, not a pass. True coverage requires a held-out calibration set the project does not have at this N; the constant-width intervals should be treated as rough, not certified.

The active-learning queue ranks unrated items by $Z(\text{uncertainty}) + Z(\text{novelty})$, where uncertainty is ensemble disagreement and novelty is kNN distance in PCA space [`active_learning_ranker.py`]. The current queue is entirely Games (Figure 16) — the sparsest, highest-uncertainty domain — which is the correct thing for the system to want labelled next.

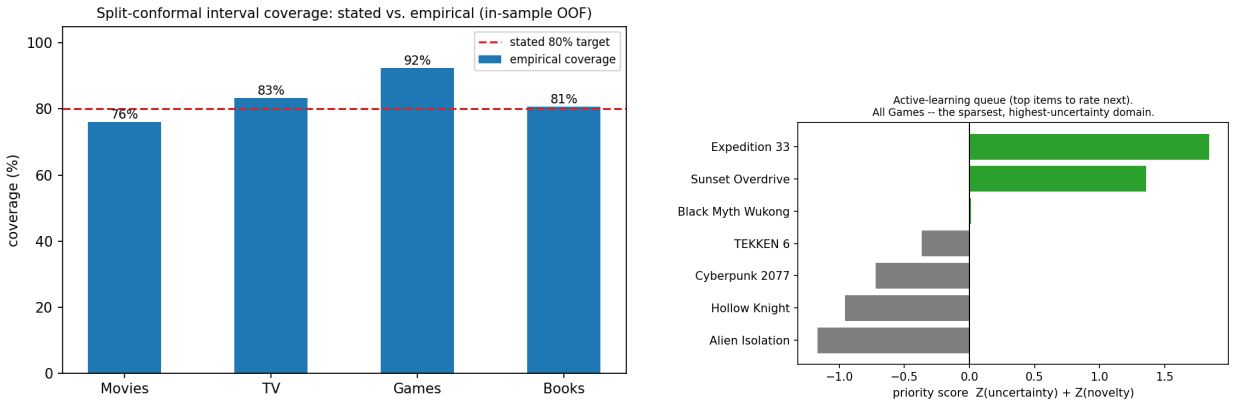


Figure 16: Left: stated vs. empirical conformal coverage per domain. Right: active-learning priority queue (uncertainty/novelty split is not persisted to the queue artifact, so the composite score is shown).

10 Evaluation Methodology and Reproducibility

Frozen-fold registry. A single `RepeatedKFold(10,5,seed=42)` = 50 folds is built once over the rated unified set and stored as a `global_id` → fold-list map [`models/unified/fold_registry.json`; `create_frozen_folds.py`]. Every experiment — benchmarks, ablation, distillation, transfer, the new edge-loss ablation — reuses it, so comparisons are paired at the item level and reproducible by construction.

Single source of truth. `comprehensive_evaluator.py` computes all benchmark/slice/ablation/distillation numbers and writes `latest_metrics.json`; `render_docs.py` renders that JSON into the README and report between markers. No metric is hand-typed into prose. This report follows the same rule — its tables are generated from the artifacts by `paper/make_figures.py`.

Failures caught and corrected (a contribution).

- **Benchmark-inversion / 5× N inflation.** An earlier writer published the unified per-domain *slice* as if it were the standalone benchmark, counting OOF rows (∼ 5× items) as N. Fixed by regenerating genuine standalone OOF deduplicated to one row per item, guarded by assertions that benchmarks ≠ slices and N = unique items [`standalone_benchmarks.py`].
- **Leak-faked significance.** The domain-identity leak (Check 4) had produced a "significant" bridge result ($p = 0.004$) that vanished once `has_*_feats` were removed [`PIPELINE_RUN_LOG.md`].
- **Stale-artifact drift.** The shipped music model expected 167 features while the regenerated feature matrix had 127, crashing the Music Oracle; the model was retrained to the current 127-feature space and a feature-count guard was added [`src/music/model_training.py`]. A concrete instance of the drift the single-source-of-truth discipline exists to prevent.
- **Superseded pilots.** The 6-fold transfer pilot ([**NULL/DROPPED**]) and the capped 1,173-link bridge run are retained in the log but explicitly marked superseded; only the latest 50-fold / 2,616-link numbers are used.

11 Limitations

- **Aleatoric ceiling, unquantified.** Low R^2 in TV/Books is partly irreducible single-user noise, but the ceiling is not estimated ([**NOT FOUND IN ARTIFACTS — ceiling value**]).
- **Single user.** $N = 1$; nothing here claims to generalise across people.

- **Small-N domains.** Games/Books/TV have $N < 200$; significance is fragile and the transfer reversal hinged on getting full statistical power.
- **Music has no ground-truth stars.** The music label is a Positive-Unlabeled affinity learned from implicit feedback (not a stated rating), so the music line is kept strictly secondary and never enters the frozen-fold taste metrics.
- **Underpowered bridges.** The entity-bridge effect is a non-significant nudge (ΔMAE $+0.019$, $p = 0.077$) and the adaptation slice is degenerate ($N = 22$).

12 Conclusion and Future Work

PMIH establishes, on honest frozen-fold evaluation, that single-user taste *can* transfer into a data-poor domain (Games) from richer ones (Movies), that this survives four adversarial controls — a label-permutation placebo among them — and that feature-space geometry shows *no detectable link* (inconclusive at $n = 12$) to which transfers work. It establishes equally clearly what does *not* help: Ridge stacking and residual heads (over-fit at small N), the distillation prior, and — newly — the custom edge-penalty loss at its tuned operating point.

Future work. (1) **YouTube as a sixth, implicit-feedback domain**: watch-history is ingested and enriched [`src/youtube/`] but deliberately *not* modelled — watch-time is a weak, confounded label and including it now would import noise into the clean five-domain transfer story; it is scoped for a separate implicit-feedback study. (2) **Full-grid confirmations** for the borderline source configs that the controls weakened. (3) **The censoring experiment**, once enough movie↔book adaptation pairs are rated to make it powered. (4) **Production text normalization**, promoting the control fix into the live embedding pipeline.

A Appendix A: Figures and Tables Generated

All regenerated by `python -m paper.make_figures` and `python -m src.experiments.edge_loss_ablation`. Files in `paper/figures/` and `paper/tables/`.

Artifact	Source data
`rating_distributions.png`	`training_features.csv`
`feature_coverage_heatmap.png`	`training_features.csv`
`pred_vs_actual_per_domain.png`	`standalone_oof.csv`
`calibration_per_domain.png`	`standalone_oof.csv`, conformal widths
`r2_achieved_per_domain.png`	`latest_metrics.json` (ceiling omitted)
`slice_vs_standalone.png`	`latest_metrics.json`
`transfer_affinity_heatmap.png`	`transfer_grid_summary.json`
`transfer_learning_curves.png`	`transfer_grid_summary.json`
`games_forest_plot.png`	`transfer_grid_summary.json`
`similarity_vs_transfer.png`	`transfer_grid_summary.json` + recomputed centroids
`transfer_controls_narrowing.png`	`transfer_grid_summary.json`
`umap_domain_and_rating.png`	`training_features.csv` (UMAP)
`coral_separability.png`	`PIPELINE_RUN_LOG.md` Check 3
`text_length_per_domain.png`	`PIPELINE_RUN_LOG.md` Check 2 (medians)
`entity_link_counts.png`	`entity_links.csv`
`conformal_coverage.png`	`standalone_oof.csv`
`active_learning_queue.png`	`ACTIVE_LEARNING_QUEUE.md`
`edge_loss_pred_vs_actual_3panel.png`	`edge_loss_ablation_results.json`
`edge_loss_calibration.png`	edge-loss ablation
`edge_loss_alpha_sweep.png`	edge-loss ablation

censoring_asymmetry	*NOT generated — only 2 movie→book adaptations*
`per_domain_summary.tex`	`training_features.csv`
`benchmark.tex`	`latest_metrics.json`, `production_oof.csv`
`edge_loss.tex`, `alpha_sweep.tex`	`edge_loss_ablation_results.json`
`architecture_ablation.tex`	`latest_metrics.json`
`distillation.tex`	`distillation_ablation_results.json`
`transfer_positive.tex`	`transfer_grid_summary.json`

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